Myo Minn **Oo**

MBBS, PhD

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As an epidemiologist, I am passionate about global health, addressing health disparities in underserved populations, and conducting implementation research. I am particularly interested in applying data science, automation, and data products to real-world health research challenges. In fact, even this CV was compiled using code and a data-driven approach.

Education

Prince of Songkla University — *Hat Yai, Thailand*

PhD, Epidemiology — *Aug 2018 – Jul 2021*

- Dissertation: "Myanmar Migrants with Tuberculosis: Health Insurance, Delays in Access to Healthcare, Model of Care on Treatment Outcomes and Quality of Life in a Border Area of Thailand and Myanmar"
- Committee: François Henry Nosten (Chair), Tippawan Liabsuetrakul (Advisor), Virasakdi Chongsuvivatwong, Sawitri Assanangkornchai, Wit Wichaidit
- Received an excellent evaluation after successfully defending my PhD thesis.

The Union, MSF Luxembourg, WHO-TDR — *Nepal & Sri Lanka*

Structural Operational Research IniTiative (SORT-IT) Training Course — *Mar 2016 – Dec 2016*

- Research project: "Alarming attrition rates among HIV-infected individuals in pre-antiretroviral therapy care in Myanmar, 2011–2014" (doi: 10.3402/gha.v9.31280)

University of Medicine — *Mandalay, Myanmar*

MBBS — *Jan 2007 – Dec 2012*

- Produced a report on vital statistics for a township in Myanmar that is among the most under-developed and remote areas of the country

Research Interests

Underserved Populations (Sex workers, gbMSM, Migrants, Indigenous Peoples), Infectious Disease Epidemiology, TB, STBBI, Global Health, Health Service Research, Applied data science in health.

Professional Experience

Waapihk Research — *Winnipeg, Canada*

Epidemiologist — *October 2023 —- Present*

- Delving into the field of indigenous health and health economics in Canada

University of Manitoba — *Winnipeg, Canada*

Post-doctoral Fellow — *May 2022 —- October 2023*

- I led the epidemiological analyses of projects in Kenya that aim to enhance understanding of the epidemiology and transmission dynamics of HPV in female sex workers and men who have sex with men (MSM).
- I also contributed to creating bioinformatics pipelines, specifically focusing on pipelines for bulk and single-cell RNA sequencing, and trained PhD students on these workflows.
- Initiated collaboration with PHDA and HOYMAS to start research projects in epidemiology aimed at gaining a better understanding of health service access, utilization patterns, mortality rates, and maternal and child health outcomes among sex workers, including the MSM population.
- Developed interactive Shiny projects and automated dashboards for improved monitoring and communication purposes.
- This work resulted in a peer-reviewed paper, two manuscripts are under writing, and a successful co-authored grant.

PLoS ONE

Academic Editor — *Jan 2022 —- Present*

- Serving as an academic editor, overseeing the peer review process, making editorial decisions, and ensuring that the submissions met the standards of the journal.

Swiss Tropical and Public Health Institute (University of Basel) — *Basel, Switzerland*

Post-doctoral Scientific Collaborator — *Aug 2021 —- Apr 2022*

- Conducted a feasibility study to assess the accuracy, reliability, and viability of calculating under-five mortality through repeated malaria surveys.
- This work resulted in a manuscript (under writing) and a completed technical report that provides evidence and insights on malaria preparedness in Papua New Guinea.

Papua New Guinea Institute of Medical Research
Goroka, Papua New Guinea

Senior Research Fellow — *Aug 2021 —- Apr 2022*

- Successfully oversaw the National Health Facility Survey of 2021 and ensured its timely and effective completion.
- Designed and developed an automated interactive dashboard to facilitate the monitoring of the survey in a streamlined and efficient manner.
- Delivered technical consultations to both the Institute and the National Technical Working Group for Malaria.
- Completed a draft for an automated report for the 2021 Health Facility Survey.
- Conceptualized a QR linkage system for implementation in the upcoming malaria indicator survey.

Prince of Songkla University
Hat Yai, Thailand

Doctoral Student (Career Break) — *Aug 2018 —- Jul 2021*

- Earned a full doctoral scholarship and secured two research grants to support my PhD field project.
- Published five first-authored papers, presented at the CDC Migration Summit in Singapore, and received the best oral presentation award in Indonesia.
- This work produced three important papers that offer valuable evidence to enhance health insurance coverage and healthcare access for migrants in Thailand, providing policymakers with insights and recommendations to address the challenges faced by these populations.

The Union, South-East Asia Office
Mandalay, Myanmar

Research Fellow & Technical Aide (based in Myanmar) — *Aug 2016 —- Jul 2018*

- Undertook over 20 research projects across various countries, including Myanmar, Bhutan, Nepal, India, and China.
- Supervised and mentored over 10 participants
- Generated over 25 research outputs and publications in peer-reviewed international journals.
- Organized and led the third National SORT-IT course, as well as a workshop on TB research agenda in Myanmar.

The Union, Country Office
Mandalay, Myanmar

Monitoring & Evaluation Officer/Manager — *May 2014 —- Jul 2016*

- Completed a one-year SORT-IT training program with success and published my first paper as the primary author, despite facing difficult health conditions and personal challenges during that period.
- Spearheaded the Monitoring and Evaluation Unit responsible for maintaining accurate records for more than 35,000 patients.
- Examined the attrition rates of the program and assessed the existing tracing system, resulting in subsequent improvements and more effective call-back procedures.
- Enhanced daily oversight of a team of 15 or more employees to promote streamlined workflows and enhanced communication.

The Union, Country Office
Mandalay, Myanmar

Medical Officer — *Feb 2014 —- Dec 2015*

- Conducted daily outpatient consultations for over 50 patients with TB/HIV.
- Provided medical care and support to patients, which included prescribing essential treatments such as anti-retroviral drugs.
- Conducted an analysis and authored my inaugural abstract on the survival analysis of patients with HIV.

Grants, Awards, and Honors

Thailand Education Hub for ASEAN Countries (THE-AC)
Thailand

Full Doctoral Scholarship — *Aug 2018 - Jul 2021*

- Awarded a complete three-year scholarship to pursue a Doctoral degree in Epidemiology.
- Produced a technical report and policy document to inform the Thai government on healthcare access issues for marginalized populations, and organized a dissemination meeting with local, regional, and international stakeholders to share our findings and recommendations.

Health System Research Institute (HSRI), Thailand
Thailand

Research Grant (Grant number: 62–082) — *Jul 2019 - Jan 2021*

- Awarded to conduct research and produce policy documents on the topics of health insurance, legal status, and their effects on the health outcomes of Myanmar migrants in Thailand.
- Funding amount: $20,000 USD

Fogarty International Center, National Institutes of Health, US
Thailand

Research Grant (Grant number: D43TW009522) — *Jul 2019 - Jan 2021*

- Awarded to conduct fieldwork for my PhD in the border area between Thailand and Myanmar.
- Funding amount: $4,000 USD

13th International Post Graduate Forum on Health System and Policy
Yogyakarta, Indonesia

Best Oral Presentation Award — *Jul 2019*

- Prize: $100 USD

The International AIDS Society
The IAS Conferences

Abstract Mentorship — *2016 - Present*

- Providing mentoring to improve the quality of abstracts and increase their chance of acceptance.

Center for Operational Research, The Union
Myanmar

Research Fellowship — *Aug 2016 - Jul 2018*

- Facilitating evidence-based policy changes by conducting operational research.

The Union, MSF Luxembourg, & WHO-TDR *Sri Lanka, China, Myanmar*

MENTORSHIP *Mar 2016 - Jul 2018*

- Supervised over 10 participants and collaborated on more than 25 publications that have contributed to the advancement of evidence-based policy changes in South Asia through the implementation of operational research.

Selected Publications

Since 2016, I have authored **38 research papers**. Some highlights are listed below, with citations taken from Google Scholar on 9 January 2024. My **h-index** is **13** with total citations of 495.

For a complete list of publications, see *my Google Scholar profile*.

1. Oo, M. M., Moore, S., Gibbons, S., Adhiambo, W., Muthoga, P., Siele, N., Akolo, M., Gebrebrhan, H., Sivro, A., Ball, B. T., et al. (2023). High prevalence of vaccine-preventable anal human papillomavirus infections is associated with HIV infection among gay, bisexual, and men who have sex with men in nairobi, kenya. *Cancer Medicine*.
2. Oo, M., Liabsuetrakul, T., Boonathapat, N., Aung, H., & Pungrassami, P. (2022). Treatment success among myanmar migrants with TB in thailand. *The International Journal of Tuberculosis and Lung Disease*, *26*(6), 550–557.
3. Oo, M. M., Boonathapat, N., Aung, H. K. K., Pungrassami, P., & Liabsuetrakul, T. (2021). Changes in health-related quality of life and the associated factors among myanmar migrants with tuberculosis: A cohort study. *BMC Infectious Diseases*, *21*(1), 1–8.
4. Feng, N., Edwards, J. K., Owiti, P. O., Zhang, G.-M., Rueda Vallejo, Z. V., Hann, K., Zhou, S.-S., Oo, M. M., Geoffroy, E. M., Ma, C., et al. (2021). Operational research capacity building through the structured operational research training initiative (SORT-IT) in china: Implementation, outcomes and challenges. *Infectious Diseases of Poverty*, *10*(1), 1–4.
5. Oo, M. M., Liabsuetrakul, T., Boonathapat, N., Aung, H. K. K., & Pungrassami, P. (2021). Association of health insurance and documentation with stigma and social support among myanmar migrants with tuberculosis before and during thailand's policy on border closure due to COVID-19: A cross-sectional study. *Journal of Racial and Ethnic Health Disparities*, 1–13.
6. Oo, M. M., Tassanakijpanich, N., Phyu, M. H., Safira, N., Kandel, S., Chumchuen, K., Zhang, L. M., Kyu, H. A., Sriwannawit, P., Bilmumad, B., et al. (2020). Coverage of tuberculosis and diabetes mellitus screening among household contacts of tuberculosis patients: A household-based cross-sectional survey from southern thailand. *BMC Public Health*, *20*(1), 1–10.
7. Oo, M. M., Tun, N. A., Lin, X., & Lucero-Prisno III, D. E. (2020). COVID-19 in myanmar: Spread, actions and opportunities for peace and stability. *Journal of Global Health*, *10*(2).

BOOKS

1. Oo, M. M. (2020). *A beginner guide to research data collection using EpiData*. Amazon Self-Publishing. `https://www.amazon.com/Beginner-Guide-Research-collection-EpiData/dp/B08F7M67QP`
2. Oo, M. M. (2023). *Principles and application of evidence-based public health practice: Chapter (16) the need for a paradigm shift to ensure adequate skilled human resources for effective public health practice*. Under Editorial Review @ Elsevier.

R PACKAGES

1. Oo, M. M. (2020). *mStats: Medical data analysis*. `https://CRAN.R-project.org/package=mStats`
2. Oo, M. M. (2021). *u5mr: Under-five child mortality estimation*. `https://CRAN.R-project.org/package=u5mr`

Works in progress

1. Flow cytometry analysis and single cell transcriptome profiling of tissue resident memory t cells within the human cervical mucosa: Comparison of ectocervix and endocervix. (2023). In *Writing*.
2. Human papilloma virus genotypes and associated factors among female sex workers in nairobi, kenya. (2023). In *Analysis & Writing*.
3. Longitudinal trajectories of SARS-CoV-2 antibodies and incidence of breakthrough infections in a 2-year follow-up cohort. (2023). In *Analysis & Writing*.
4. Under-five child mortality estimation from repeated malaria indicator surveys in papua new guinea. (2023). In *Analysis & Writing: Vols. In collaboration with Swiss TPH and PNG IMR*.
5. Divergent HIV acquisition and inflammation outcomes associated with IFN-a2 and IFN-b expression in the female genital tract. (2023). In *Data Acquisition & Analysis*.

Projects

1. *SWOP HPV cohort with support from bill & melinda gates foundation*. (2023).
2. *COVID-19 antibody trajectory analysis project*. (2023).
3. *Patterns of healthcare access and health service utilization at SWOP clinics (data acquisition)*. (2023).
4. *Mortality and mental health among female sex workers in nairobi (preparing data collection)*. (2023).
5. *Hotspot mapping of sex workers in nairobi before and after COVID0-19 pandemic (data acquisition)*. (2023).

Data Science Products

mStats: Epidemiological Data Analysis

R

- R package for data management and epidemiological data analysis inspired by STATA

u5mr: Under-Five Child Mortality Estimation

R

- R package for estimating under-five mortality rates using indirect methods mentioned in UN X manual and papers by Rajaratnam e.t.

MalCon

R, Rmarkdown, GitHub

- A GitHub static webpage for Malaria Control Unit (MalCon) at Papua New Guinea Institute of Medical Research (PNGIMR)

MalCon Dashboard

R, Rmarkdown, Automation, CICD, ODK, Flexdashboard

- An interactive dashboard for monitoring the Health Facility Survey 2021 conducted in Papua New Guinea

Personal Budget Manager

R, Shiny, MongoDB, shinyapp.io

- A Shiny budget management app built with the Golem framework, free user registration, passcode authentication

COVID-19 Antibody Trajectory App

R, Shiny, shinyapp.io

- A Shiny data exploratory app for the purpose of communication with stakeholders, build with the Golem framework with user authentication

XpertMagic

R, Shiny, shinyapp.io

- A Shiny app to convert raw outputs from Gene Xpert machine for HPV genotypes and CTNG to tidy format, pre-specified upload data size

Personal Webpage hosted on GitHub

R, Rmarkdown, GitHub

- A GitHub static personal webpage

Personal Webpage hosted on netlify

R, Rmarkdown, blogdown, netlify

- A netfliy personal webpage

Presentations

CAHR 2023 — *Canada*

Oral Presentation — *Apr 2023*

- Single Cell Transcriptomics Delineate Divergent Localizations Of Tissue-Resident memory T Cells Across The Human Cervical Mucosa

STD/HIV/SRH Collaborative Research Group Annual Review Meeting — *Kenya*

Oral Presentation — *Jan 2022*

- HPV epidemiology and preliminary findings from SWOP HPV Study
- High prevalence of vaccine-preventable anal human papilloma virus infections isassociated with treated HIV infection among gay, bisexual, and men who have sex with men in nairobi, kenya.

14th Postgraduate Forum of Health Systems and Policy — *Virtual*

Oral Presentation — *Dec 2020*

- Health insurance and out-of-pocket expenditure from symptoms to tuberculosis diagnosis among Myanmar migrants in a border area of Thailand

51st Union World Conference on Lung Health *Virtual*

Poster Presentation *Oct 2020*

- Coverage of health insurance schemes among Myanmar migrants with tuberculosis in a border area of Thailand and its associated factors: An interim analysis

US CDC Migration Summit *Singapore*

Oral Presentation *Nov 2019*

- Cross-border TB referral system: a case study from Thailand

7th International Health Literacy Conferenc *Vietnam*

Poster Presentation *Nov 2019*

- Tuberculosis literacy and completion of household contact investigations: A household-based survey from Southern Thailand

13th Postgraduate Forum of Health Systems and Policy *Indonesia*

Oral Presentation *Jul 2019*

- Unsuccessful treatment outcomes after oral direct-acting antivirals in people with chronic hepatitis C virus infection and its associated factors

Ph.D. Candidate examination *Thailand*

Oral Presentation *Feb 2019*

- Unsuccessful treatment outcomes after oral direct-acting antivirals in people with chronic hepatitis C virus infection and its associated factors

Teaching

Swiss Tropical and Public Health Institute *Basel, Switzerland*

Invited speaker *Oct 2021*

- Shared a personal account of migration and health at the border area.

Global Public Service Academies Program (USA) *Mae Sot, Thailand*

Instructor *Oct 2019 - Feb 2020*

- Delivered two lectures on the topic of migration and public health with a focus on tuberculosis control, to high school students from Japan, Australia, Canada, and the United States.

University of Community Health *Myanmar/Burma*

Invited lecturer *Feb 2019*

- Conducted a two-day training workshop on the application of R for medical statistics.

SORT IT course *Sri Lanka, China, Myanmar*

Mentor *Mar 2016 - Jul 2018*

- Provided mentorship to more than 20 international participants.
- Taught data management and applied statistics.
- Created and delivered a lecture on the subject of Effective Research Communication through Infographics.

Services & Outreach

Academic Editor: PLoS One (2022 - Present)

Ad Hoc Reviewer: 50+ Verified Reviews in International peer-reviewed journals (2016 - Present) ● PLoS One (2016 - Present) ● BMC Health Service Research (2016 - Present) ● Public Health Action (2016 - Present) ● Global Health Research and Policy (2016 - Present) ● Journal of Health Research (2016 - Present) ● AIDS Research and Human Retroviruses (2016 - Present) ● African Journal of AIDS Research (2016 - Present) ● International Journal of STD & AIDS (2016 - Present)

Affiliate: Prince of Songkla University (2018 - 21)

Editorial Board: Public Health Challenges (2021 - Present)

Member: National Technical Working Group (Malaria) (2021 - 22) ● PNGIMR Research Network (2021 - 22) ● SORT-IT OR Network, Myanmar Society of Epidemiology (MSE), Myanmar Medical Association (MMA) (2016 - Present)

Mentor: The International AIDS Society Conferences (2016 - Present)

Myanmar R User Group (Monthly Meetup): Virtual (2022 - Present)

R for Researchers (Monthly Meetup): University of Manitoba (2022 - Present)

Reviewer: Rady Faculty of Health Sciences Graduate Studentship Program (2022)

Skills

Development: Version Control (Git) • Automation & CICD (Travis CI, GitHub Action) • Web Hosting (Netlify, GitHub) • Shiny app (Golem framework)

Language: Burmese (Native) • Lisu (Native) • English (C2) • Chinese (A2) • Thai (A2) • French (A1) • Tok Pisin (A1) • Swahili (A1)

Markup Languages: HTML • CSS • Markdown • Rmarkdown • YAML

Programming Languages: R • Python • SQL • JAVA • JavaScript • BASH

Research: Systematic Review • Epidemiology in TB (Mahidol University, 2019) • Lab Training in TB Microbiology (Mahidol University, 2019) • Cohort study design and analysis • Epidemiology in Public Health Practice Specialization (Johns Hopkins Bloomberg School of Public Health, 2020)

Statistics and Data Science: Longitudinal Data Analysis (GEE, Mixed Models) • Causal Inference (SEM, DAG) • Meta-analysis • Epidemiological Methods • Child mortality Estimation • Statistical Modelling • Egocentric Network Analysis • Survey Methods • The Data Scientist's Toolbox (The Johns Hopkins University)

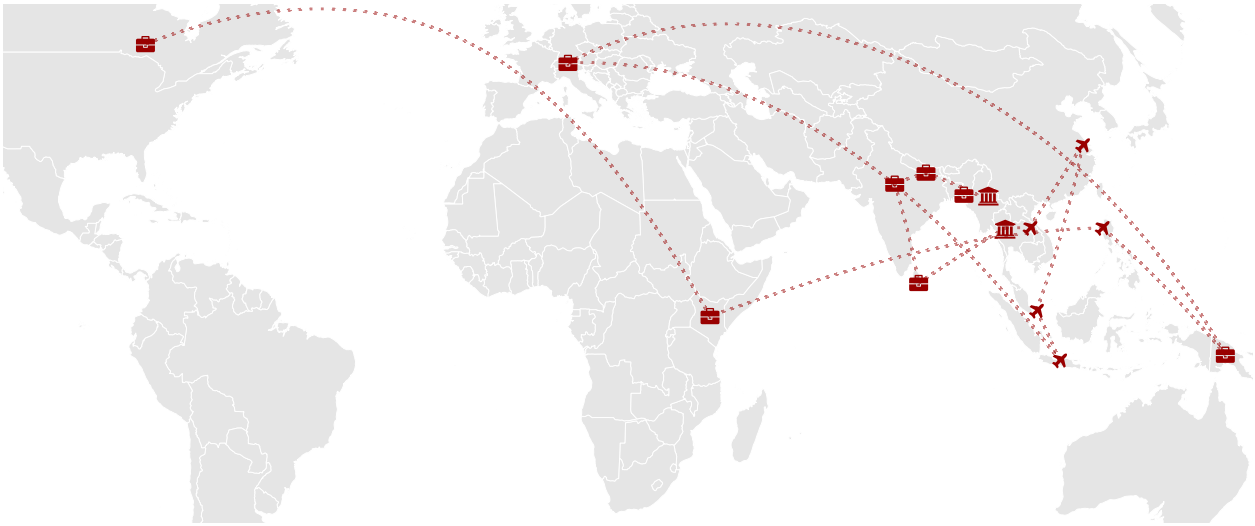
Text Editors: Rstudio • Visual Studio Code • Vim

Communication: Dashboard Design (R, Shiny, Flexdashboard) • Reproducible Research (Rmarkdown) • data visualization (ggplot2, arcGIS, qGIS) • peer-reviewed publications (journal, conference)

Online Tutorials

1. Create Publication-Ready Tables using `gtsummary`
2. R Webinars
3. Statistics & Public Health Webinars
4. Git & GitHub Basics
5. Making infographics using MS PowerPoint
6. Data entry using EpiData Manager & EntryClient
7. Introdution to R and RStudio
8. Statistics with R - ENGLISH Version
9. Statistics with R - MYANMAR Version

Charting the Course: Reflecting on My Journey Thus Far



This figure is created using the packages, `rnaturalearth`, `sf`, `sfnetworks`, `emojifont`, and `ggplot2`.

References

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Note

This document is written in `rmarkdown` and generated using `vitae` package and `awesome-cv` template. All the source code behind this `CV` is available on this GitHub repo.